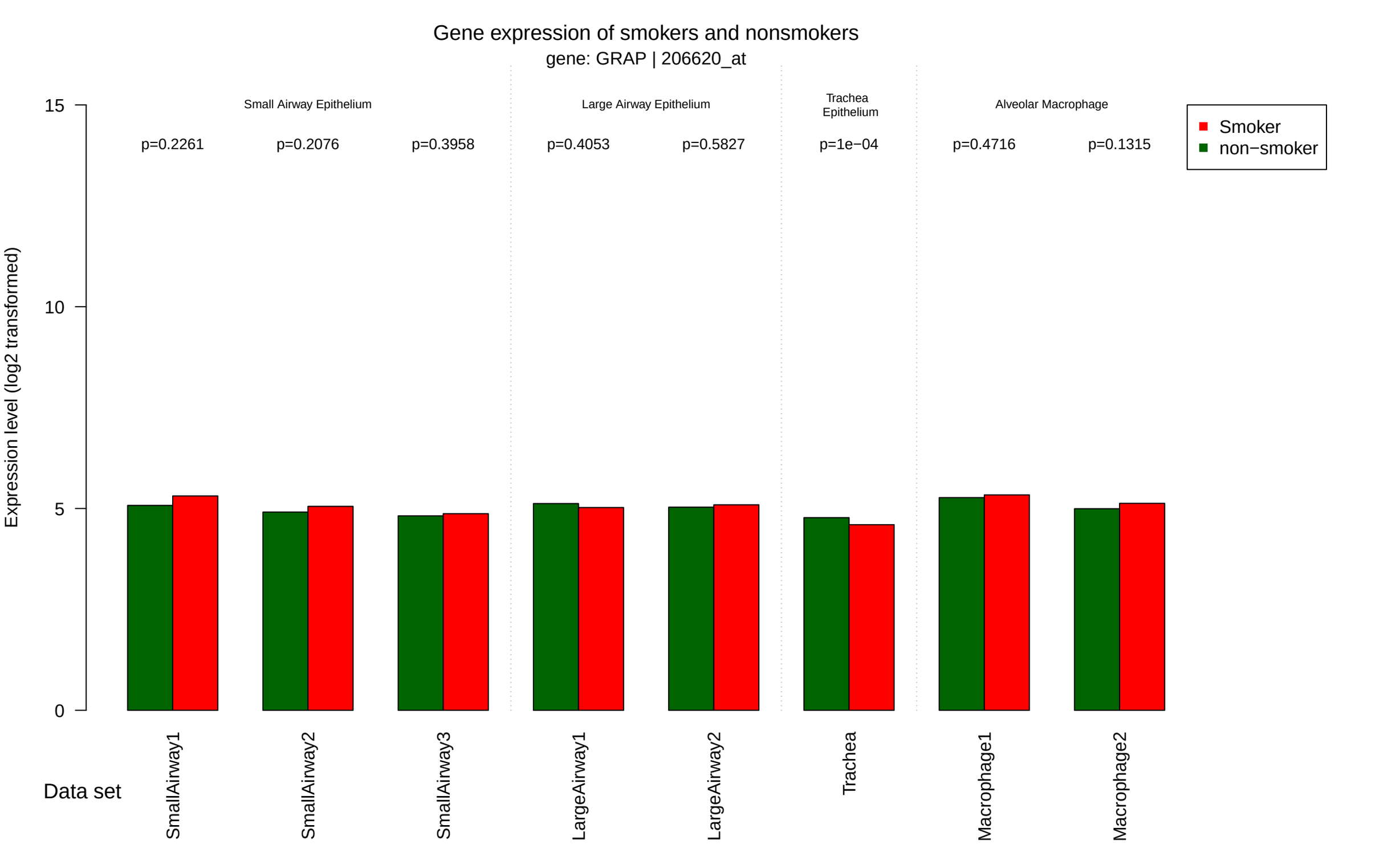
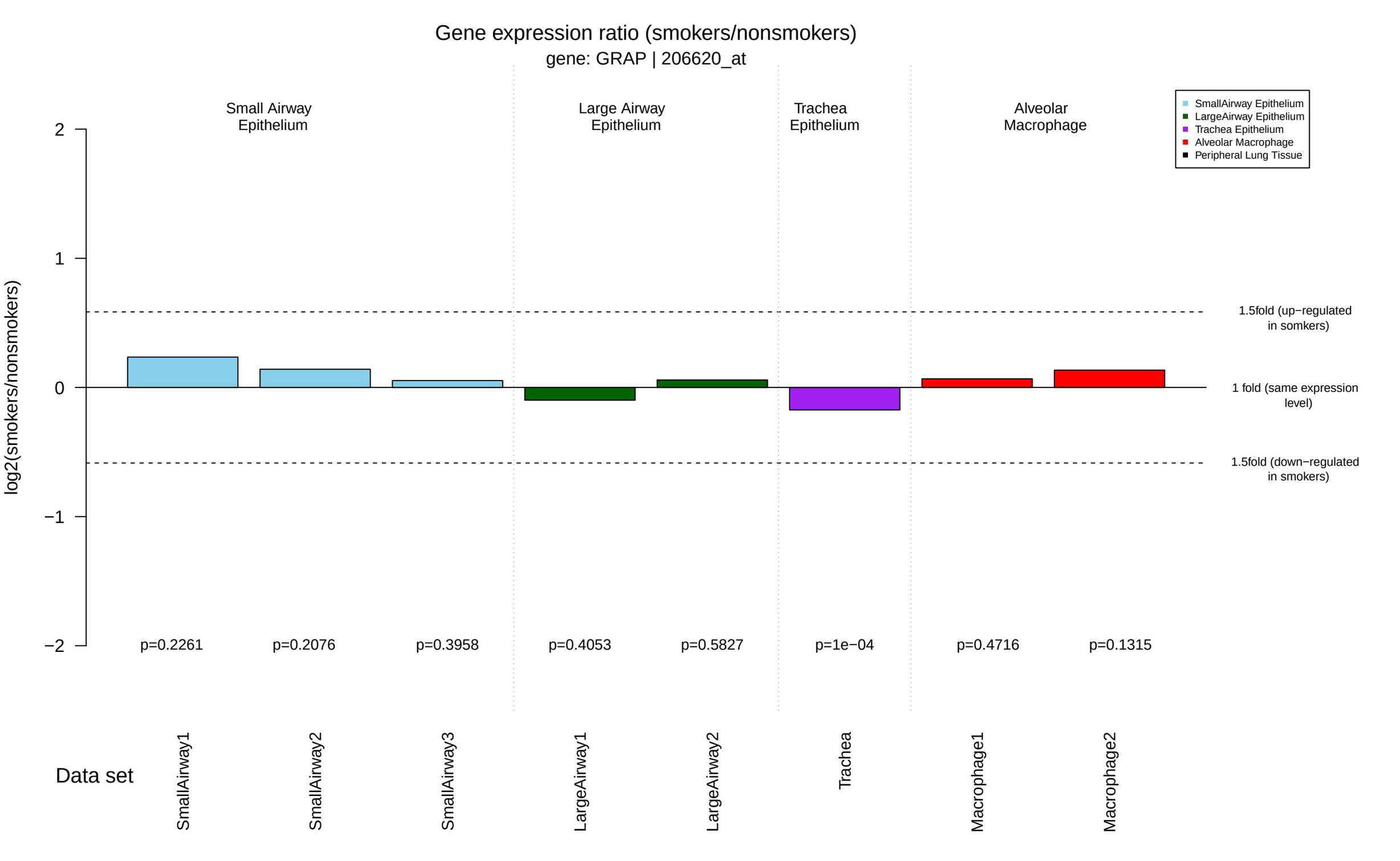
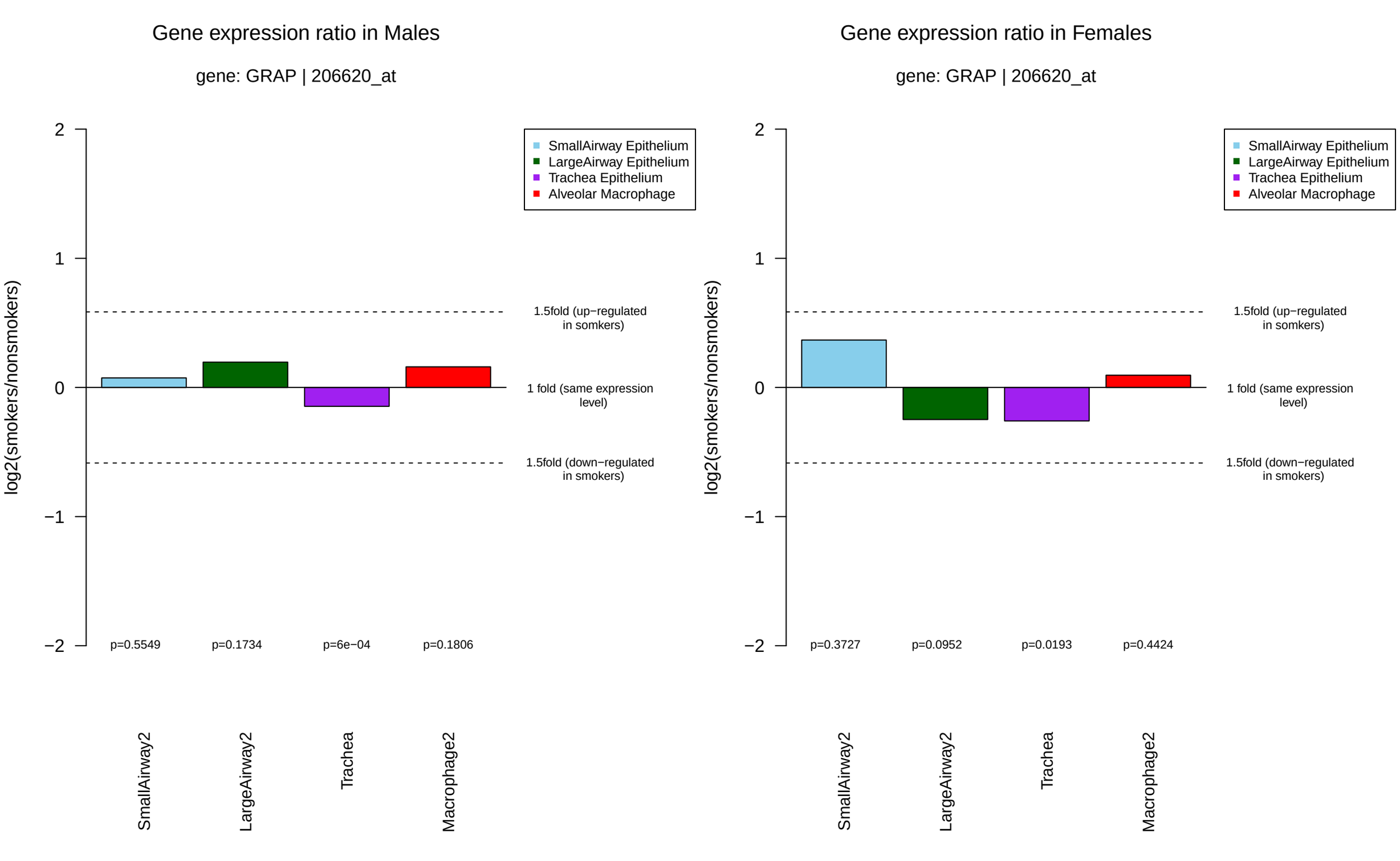
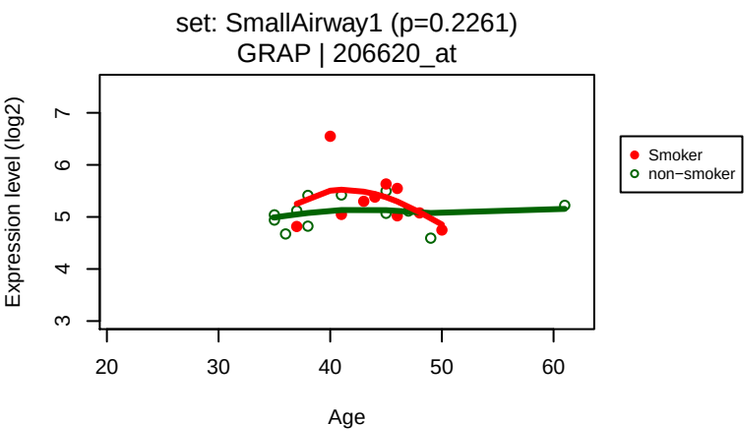
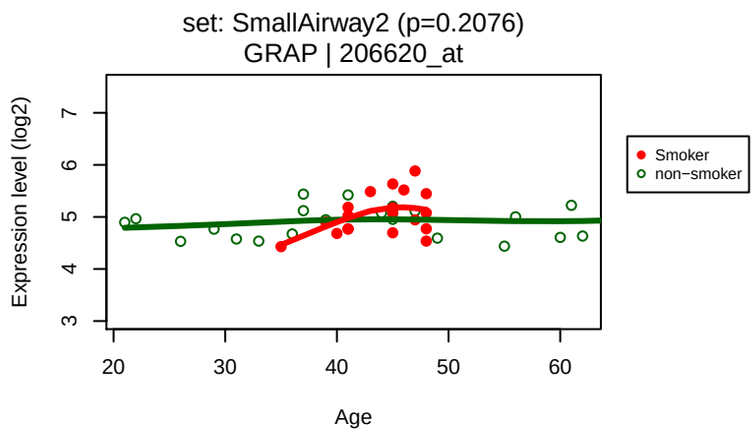
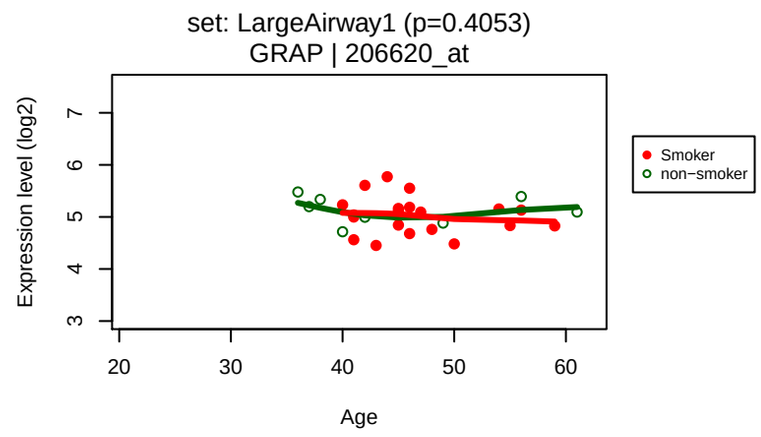

Gene expression of smokers and nonsmokers
gene: GRAP | 206620_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2261
p=0.2076
p=0.3958
p=0.4053
p=0.5827
p=1e−04
p=0.4716
p=0.1315
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

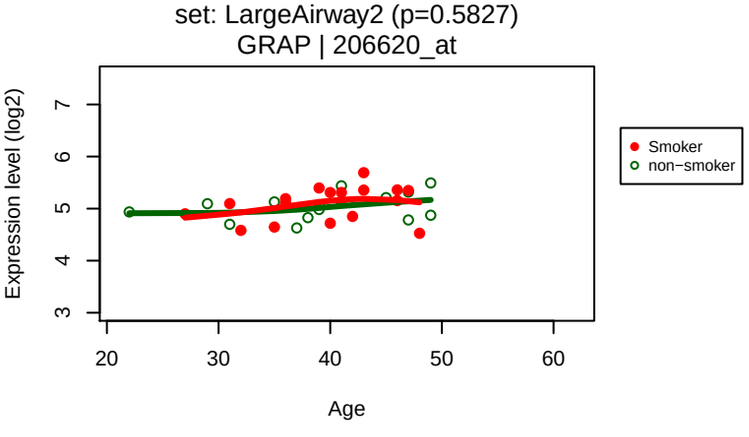

Gene expression ratio (smokers/nonsmokers)
gene: GRAP | 206620_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2261
p=0.2076
p=0.3958
p=0.4053
p=0.5827
p=1e−04
p=0.4716
p=0.1315
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

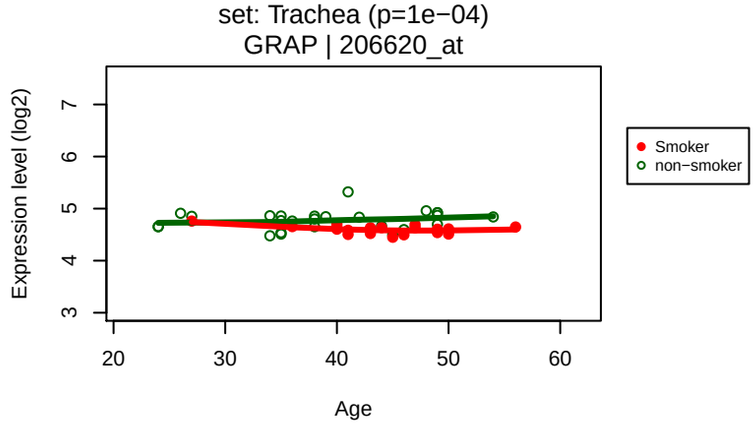

Gene expression ratio in Males
gene: GRAP | 206620_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5549
p=0.1734
p=6e−04
p=0.1806
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRAP | 206620_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3727
p=0.0952
p=0.0193
p=0.4424
SmallAirway2
LargeAirway2
Trachea
Macrophage2

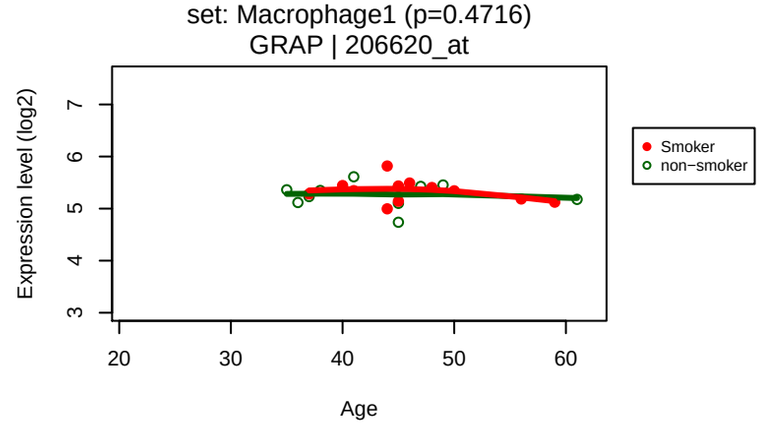

set: SmallAirway1 (p=0.2261)
GRAP | 206620_at
Expression level (log2)
Age
Smoker
non−smoker

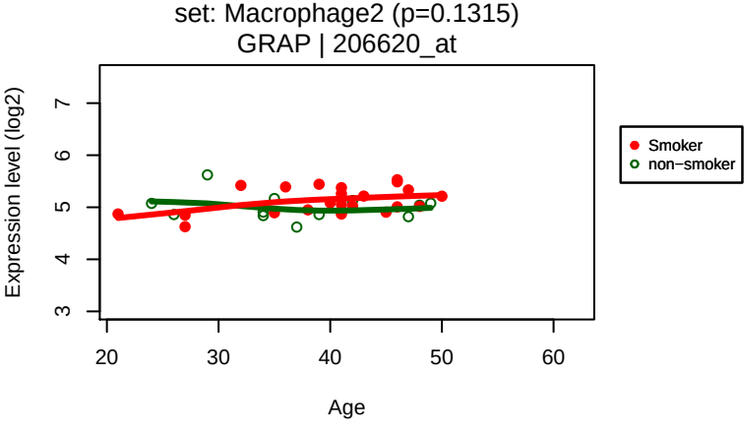

set: SmallAirway2 (p=0.2076)
GRAP | 206620_at
Expression level (log2)
Age
Smoker
non−smoker

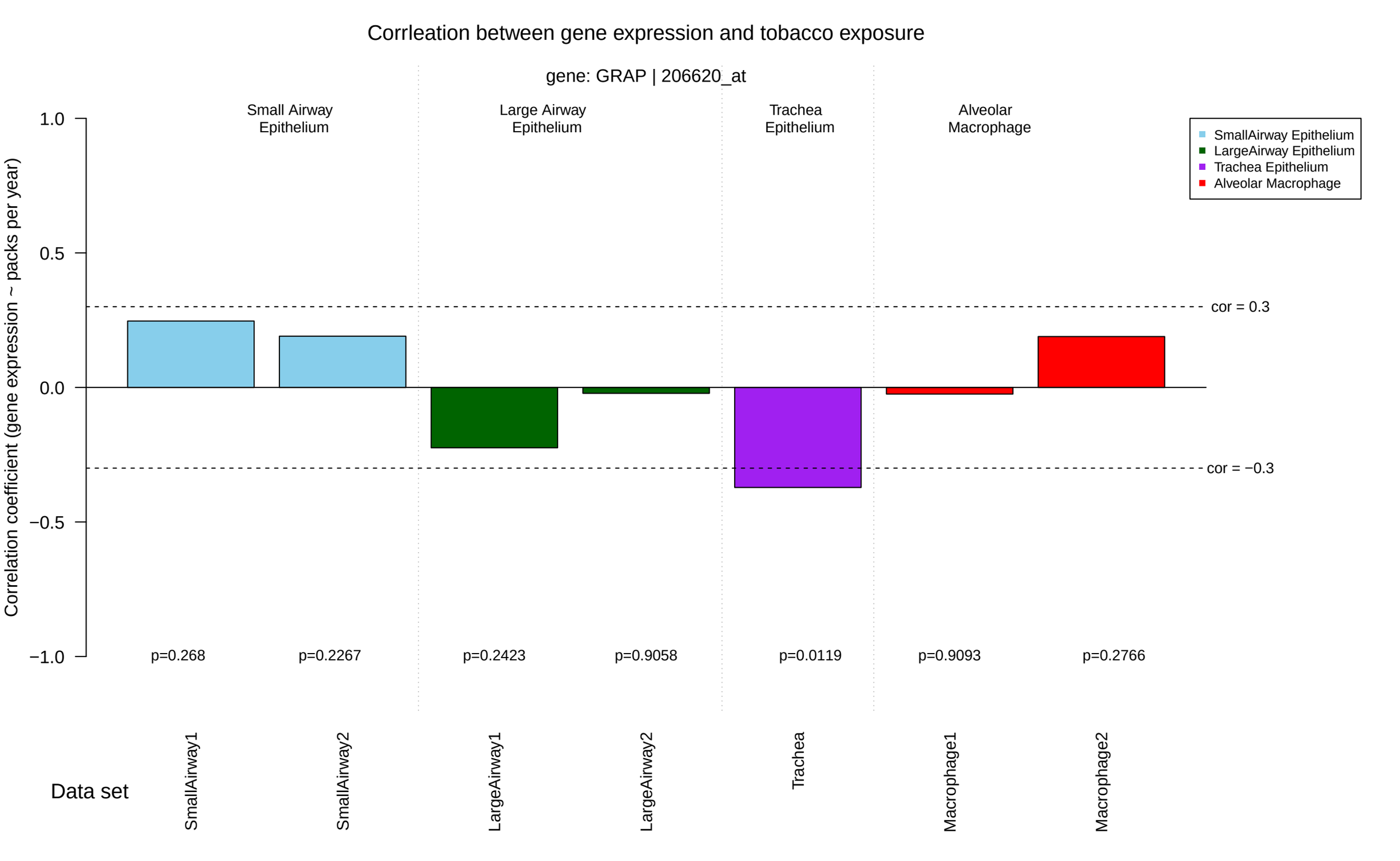

set: LargeAirway1 (p=0.4053)
GRAP | 206620_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5827)
GRAP | 206620_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
GRAP | 206620_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4716)
GRAP | 206620_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1315)
GRAP | 206620_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRAP | 206620_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.268
p=0.2267
p=0.2423
p=0.9058
p=0.0119
p=0.9093
p=0.2766
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2